

Name : Rafiur Rahman Mashrafi

ID : 2221971042

Course : CSE323

Assignment : 1

① Preemptive priority scheduling

Priority	Process	Arrival time	Burst time	Completion time	TAT	WT	RT
1	P ₁	0	5 → 3 → 1	12	12	7	0
2	P ₂	1	4 → 2 → 0	8	7	6	0
3	P ₃	2	2 → 0	4	2	0	0
4	P ₄	4	1	5	1	0	0

ready queue

P ₁	P ₂	P ₃	P ₄	P ₂	P ₁
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Gantt Chart

P ₁	P ₂	P ₃	P ₃	P ₄	P ₂	P ₁	
0	1	2	3	4	5	8	12

	Completion time	Turnaround time (CT - AT)	Waiting time (TAT - BT)	First response (start - AT)
P ₁	12	12 - 0 = 12	12 - 5 = 7	0 - 0 = 0
P ₂	8	8 - 1 = 7	7 - 4 = 3	1 - 1 = 0
P ₃	4	4 - 2 = 2	2 - 2 = 0	2 - 2 = 0
P ₄	5	5 - 4 = 1	1 - 1 = 0	4 - 4 = 0

$$\text{Avg completion time} = \frac{12 + 8 + 4 + 5}{4} = \frac{29}{4} = 7.25$$

$$\text{Avg turnaround time} = \frac{12 + 7 + 2 + 1}{4} = \frac{22}{4} = 5.50$$

$$\text{Avg waiting time} = \frac{7 + 3 + 0 + 0}{4} = \frac{10}{4} = 2.50$$

$$\text{Avg response time} = \frac{0 + 0 + 0 + 0}{4} = \frac{0}{4} = 0$$

② Round Robin Time Quantum = 2

Process	Arrival time	Burst time	Completion time	TAT	WT	RT
P ₁	0	5	12	12	7	0
P ₂	1	4	11	10	6	1
P ₃	2	2	6	4	2	2
P ₄	4	1	9	5	4	4

Ready Queue (in order)

P ₁	P ₂	P ₃	P ₁	P ₄	P ₂	P ₁
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Gantt Chart

P ₁	P ₂	P ₃	P ₁	P ₄	P ₂	P ₁	
0	2	4	6	8	9	11	12

	Completion time	Turnaround time (CT - AT)	Waiting time (TAT - BT)	Response time (start - AT)
P ₁	12	12 - 0 = 12	12 - 5 = 7	0 - 0 = 0
P ₂	11	11 - 1 = 10	10 - 4 = 6	1 - 1 = 0
P ₃	6	6 - 2 = 4	4 - 2 = 2	2 - 2 = 0
P ₄	9	9 - 4 = 5	5 - 1 = 4	4 - 4 = 0

$$\text{Avg completion time} = \frac{12 + 11 + 6 + 9}{4} = \frac{38}{4} = 9.50$$

$$\text{Avg turnaround time} = \frac{12 + 10 + 4 + 5}{4} = \frac{31}{4} = 7.75$$

$$\text{Avg waiting time} = \frac{7 + 6 + 2 + 4}{4} = \frac{19}{4} = 4.75$$

$$\text{Avg response time} = \frac{0 + 0 + 0 + 0}{4} = \frac{0}{4} = 0$$